

		Marks	Credits
L T P	Exam:	100	4
4 - -	Sessional:	50	
	Total:	150	4

Duration of Exam: 3 hrs.

Course Outcomes:

Students who complete the course will have demonstrated the ability to do the following:

- CO1 Argue the correctness of algorithms using inductive proofs and invariants.
- CO2 Analyze worst-case running times of algorithms using asymptotic analysis.
- CO3 Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it.
- CO4 Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it.
- CO5 Explain the major graph algorithms and their analyses. Employ graphs to model engineering problems

NOTE: Examiner will set nine question in total. Question One will be compulsory and will comprises of all section and remaining eight questions to be set by taking two questions from each unit. The students have to attempt five questions in total, first being compulsory and selecting one from each Unit.

UNIT 1

Foundation & Data Structure:

Foundation & Elementary Data Structure: Algorithms, Performance analysis: Space & time complexity, Growth of functions, Divide & Conquer, Recurrence Equations, Basic elements of data structure like Stacks & Queues, Trees, Graphs, Linked List, Sorting & Order statistics. Data Structure: Dynamic sets & searching: Introduction, Array doubling, Amortized time analysis, R-B trees, Hashing, Dynamic equivalence relations & Union-Find programs, Priority queues with a decrease key operation.

Graph & graph traversals: DFS, strongly connected components, Bi-connected components.

UNIT 2

Advanced Design & Analysis Techniques:

Greedy & Dynamic Method: General methods, Knapsack problem, Job sequencing with deadlines, Minimum cost spanning trees, Optimal merge patters, Single-source shortest path, 0/1 Knapsack, Multistage graphs, All-pair shortest path, Optimal binary search trees, Travelling salesperson problem, Flow shop scheduling.

Backtracking & Branch and Bound: General methods, 8 Queens problem, Sum of subsets, Graph coloring, Hamiltonian cycles, Knapsack problem, Travelling salesperson problem, Efficiency consideration.

UNIT 3

NP-Hard & NP-Complete Problems: Basic concepts, Cook's Theorem, NP-hard graph problem, NP-Hard scheduling problems.

String Matching: Introduction, A straight forward solution, The Knuth-Morris-Pratt algorithm, The Boyer-Moore algorithm, approximate string matching.

UNIT 4

Parallel Algorithms: Introduction, Parallelism, The PRAM, and other models, some simple PRAM algorithms, Handling write conflicts, Merge and Sorting, Finding connected components.

Approximation algorithms: Introduction, Absolute approximations, ϵ - approximations, Polynomial time approximation schemes, Fully Polynomial time approximations schemes.

References:

1. Computer Algorithms: Introduction to design and analysis (3rd edition) by Sara Baase and Allen Van Gelder, Pearson, 2000.
2. Fundamentals of Algorithms by Gilles Brassard and Paul Bratley
3. Design and Analysis of Algorithms (Computer science Series) by Jeffrey D. Smith Publ.
4. Fundamentals of Computer algorithms, Ellis Horowitz and SratajSahnim 1978, Galgotia publ.
5. Algorithms Design (PIE) by Eva Tardos and Jon Klienberg, person.
6. Introduction to Algorithms, Thomas h Cormen, Harles E leiseron and Ronald Lrivest : 1990, TMH.